

Advanced (Habanero)

Q1. Factor $x^2 - 4$

- (A) $(x - 2)(x + 2)$
- (B) $(x - 2)(x - 2)$
- (C) $(x + 2)(x + 2)$
- (D) $x^2 - 2$
- (E) $x^2 + 4$

Q2. Factor $9y^2 - 25$

- (A) $(3y - 5)(3y + 5)$
- (B) $(3y - 5)(3y - 5)$
- (C) $(3y + 5)(3y + 5)$
- (D) $9y^2 - 5$
- (E) $9y^2 + 25$

Q3. Factor $x^2 - 9y^2$

- (A) $(x - 3y)(x + 3y)$
- (B) $(x - 3y)(x - 3y)$
- (C) $(x + 3y)(x + 3y)$
- (D) $x^2 - 3y$
- (E) $x^2 + 9y^2$

Q4. Factor $16x^2 - 49y^2$

- (A) $(4x - 7y)(4x + 7y)$
- (B) $(4x - 7y)(4x - 7y)$
- (C) $(4x + 7y)(4x + 7y)$
- (D) $16x^2 - 7y$
- (E) $16x^2 + 49y^2$

Q5. Solve for x in $x^2 - 16 = 0$

- (A) $x = 4$
- (B) $x = -4$
- (C) $x = 0$
- (D) $x = 16$
- (E) $x = -16$

Q6. Solve for y in $y^2 - 9 = 0$

- (A) $y = 3$
- (B) $y = -3$
- (C) $y = 0$
- (D) $y = 9$
- (E) $y = -9$

Q7. Factor $x^2 - 2x - 15$ using difference of squares

- (A) $(x - 5)(x + 3)$
- (B) $(x - 5)(x - 3)$
- (C) $(x + 5)(x + 3)$
- (D) $x^2 - 2x - 3$
- (E) $x^2 - 2x + 15$

Q8. Which of the following is a difference of squares?

- (A) $x^2 + 4$
- (B) $x^2 - 4$
- (C) $x^2 - 2x + 1$
- (D) $x^2 + 2x + 1$
- (E) $x^2 - 2x - 1$

Open Ended Questions (FRQ)

Q9. Factor the expression $x^2 - 36y^2$ and explain why it is a difference of squares.

Q10. Solve for x and y in the system of equations: $x^2 - y^2 = 9$ and $x + y = 5$. Show all steps and explain your reasoning.

Chef's Tips

- Review the difference of squares formula before the test
- Emphasize the importance of factoring expressions correctly
- Provide examples of real-world applications of difference of squares